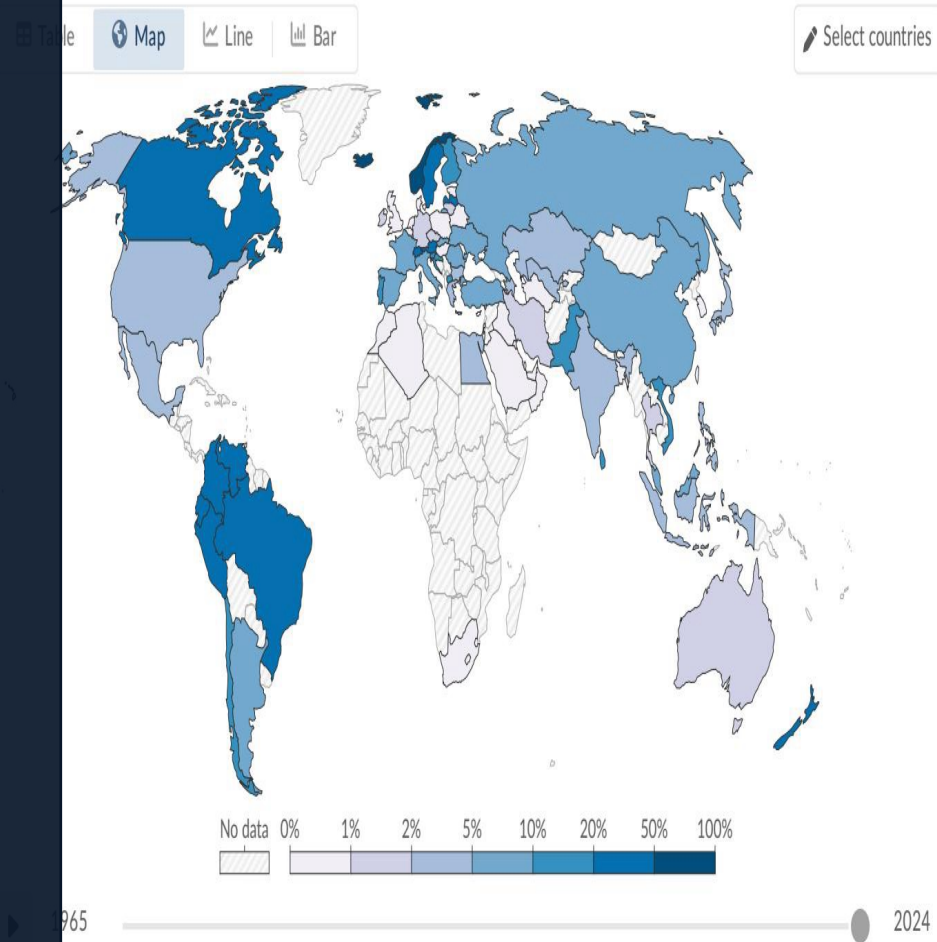


Global Renewable Energy in 2025: How Far Have We Come?

Policy analysts and energy sector strategists briefing deck

Core question: Are renewables scaling fast enough to replace fossil fuels?

Source: Our World in Data (Ember 2026; Energy Institute 2025)



Data source: Energy Institute - Statistical Review of World Energy (2025) - [Learn more about this data](#)

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Fossil Fuels Drive ~75% of GHG Emissions and Major Health Harm

The transition is both a climate and public-health imperative

~75%

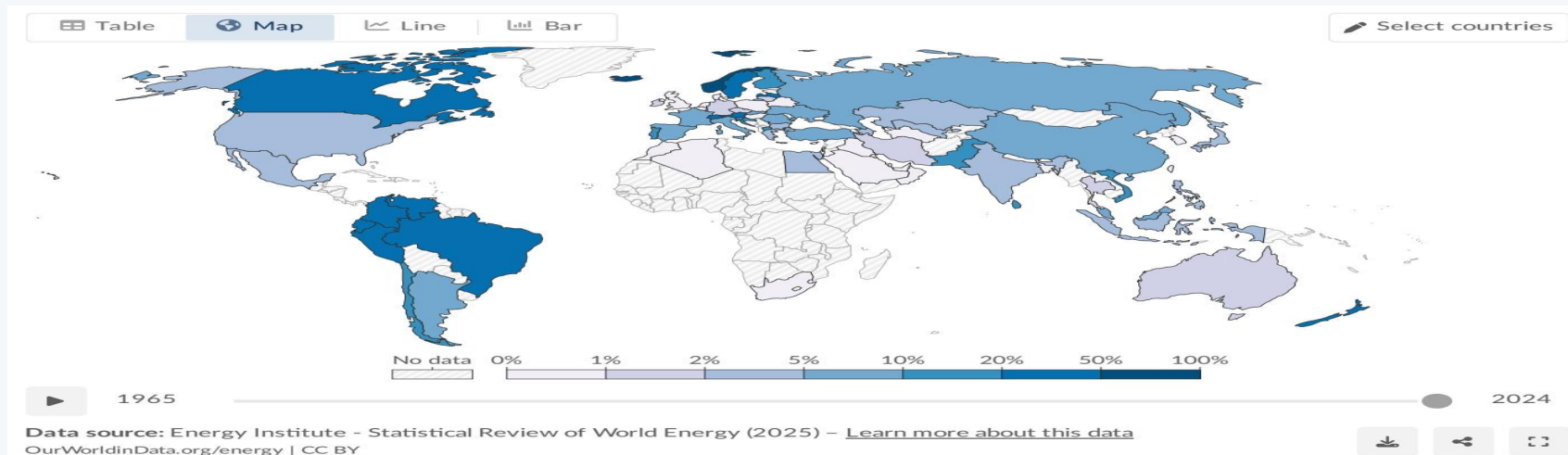
of global GHG emissions
come from energy fossil fuel
use

≥5M

premature deaths/year
linked to air pollution

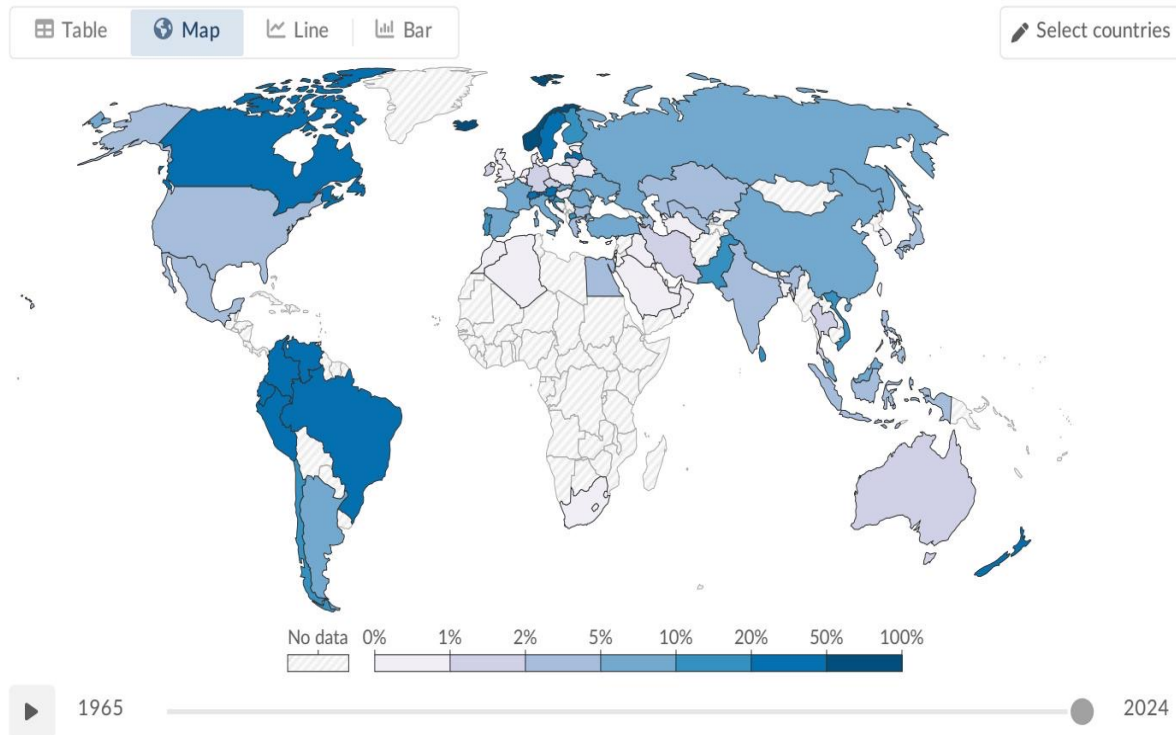
Implication

Rapid scale-up of low-carbon
electricity and end-use
electrification is non-negotiable.



Renewables Are ~14% of Global Primary Energy (2024)

Primary energy includes electricity, transport, and heating



~14% global primary energy from renewables

Why lower than electricity?

Transport and heating remain heavily oil and gas dependent, slowing whole-system decarbonization.

Regional leaders include Norway, Brazil, New Zealand; several Middle Eastern producers remain below ~5%.

Data source: Energy Institute - Statistical Review of World Energy (2025) - [Learn more about this data](#)

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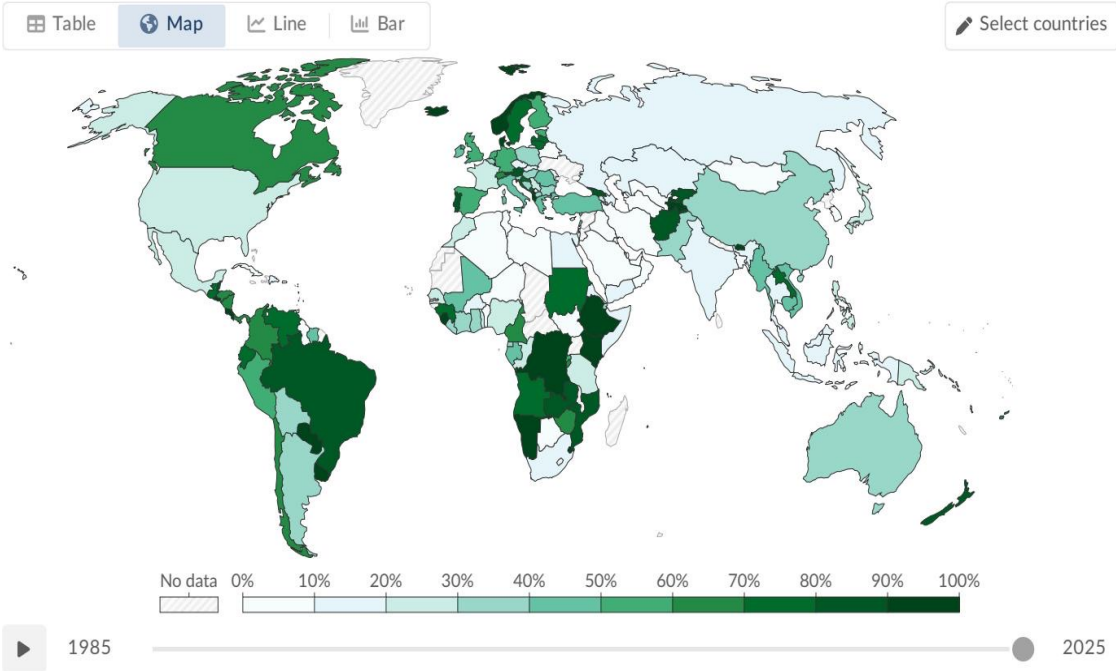
Renewables Supply Almost One-Third of Global Electricity (2025)

Electricity is the leading edge of the transition

Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.

Our World
in Data



Country/Region	Renewable Share of Electricity
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%

Range from <2% to ~98% shows structural disparities in resource endowment, policy, and investment.

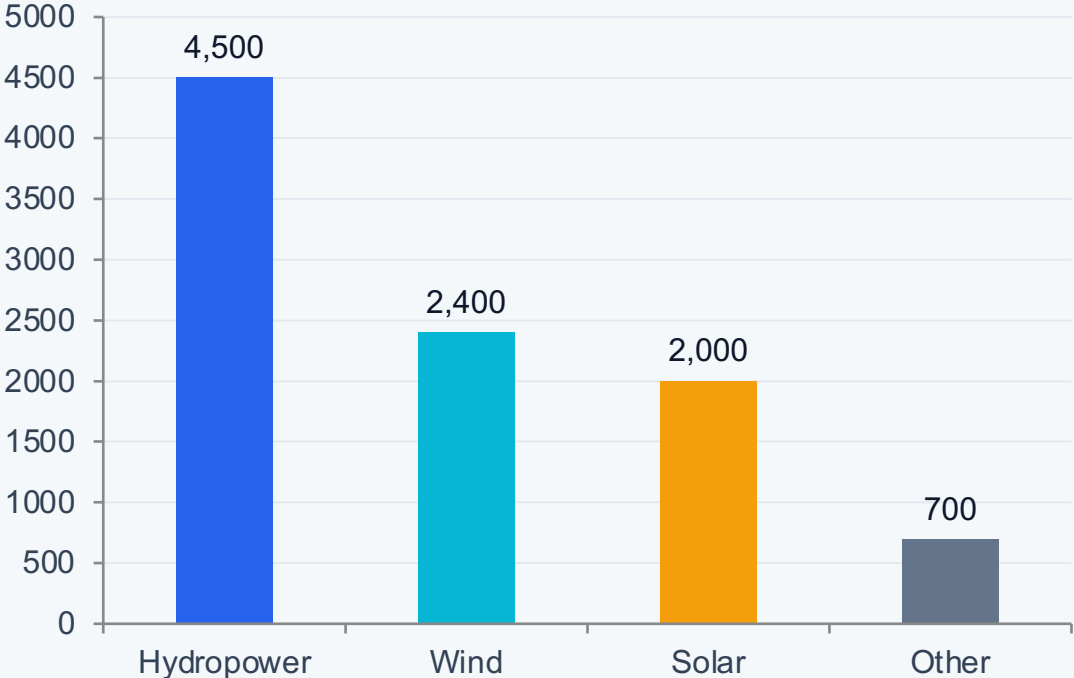
Data source: Ember (2026); Energy Institute - Statistical Review of World Energy (2025) - [Learn more about this data](#)

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Hydropower Still Leads Renewable Electricity at ~4,500 TWh

But wind and solar are closing the gap quickly

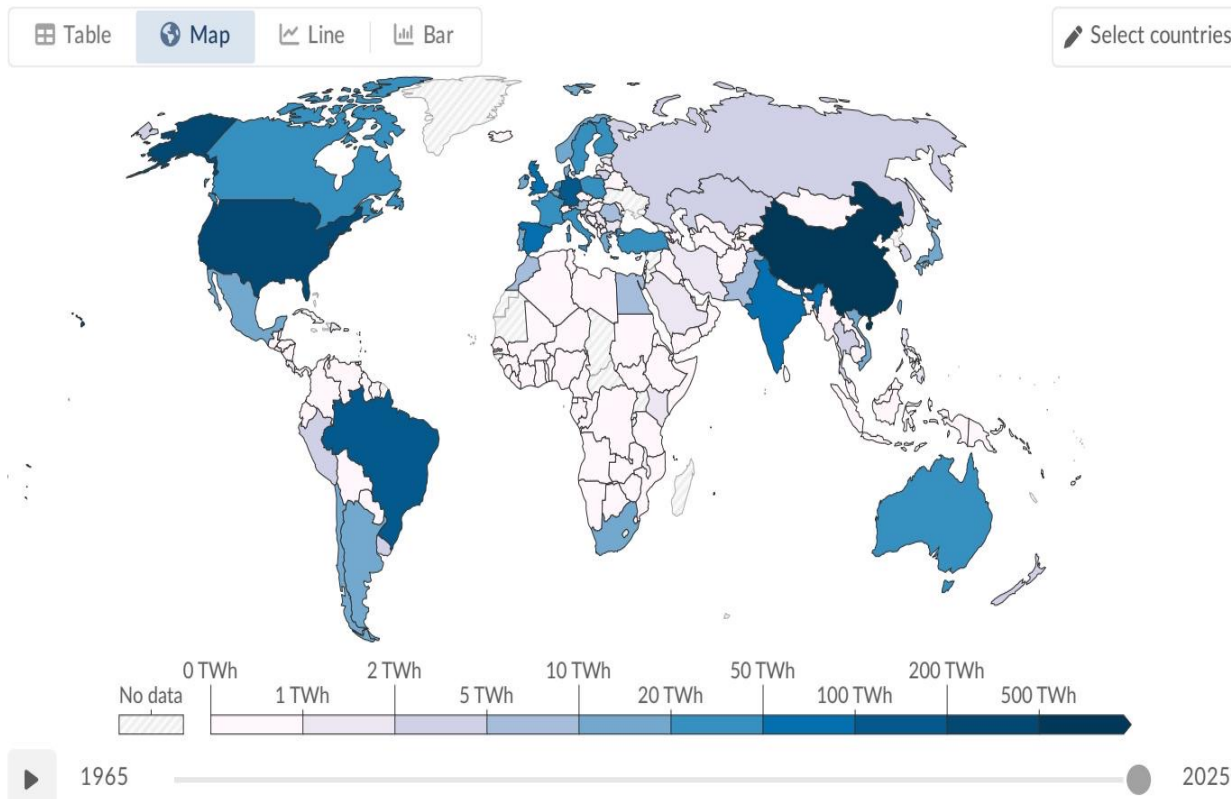


Technology	Generation (TWh)	Share
Hydropower	~4,500	~47%
Wind	~2,400	~25%
Solar	~2,000	~21%
Other	~700	~7%

Wind + solar together (~4,400 TWh) now nearly match hydropower output.

Hydropower Remains a Foundation, Concentrated in a Few Major Producers

Largest output in China, Brazil, and Canada



~47
%

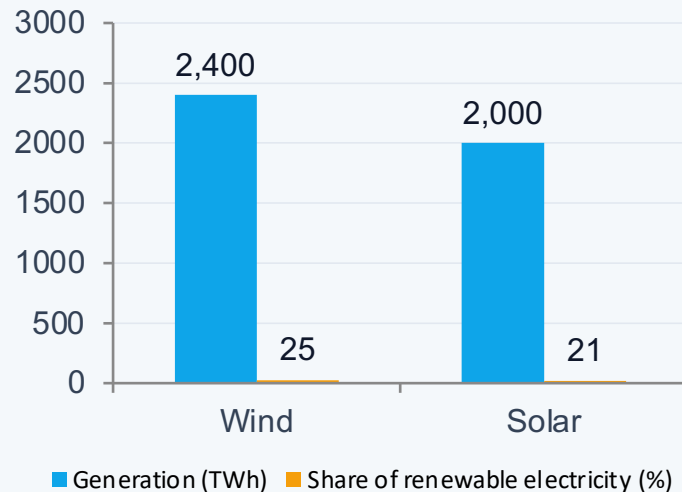
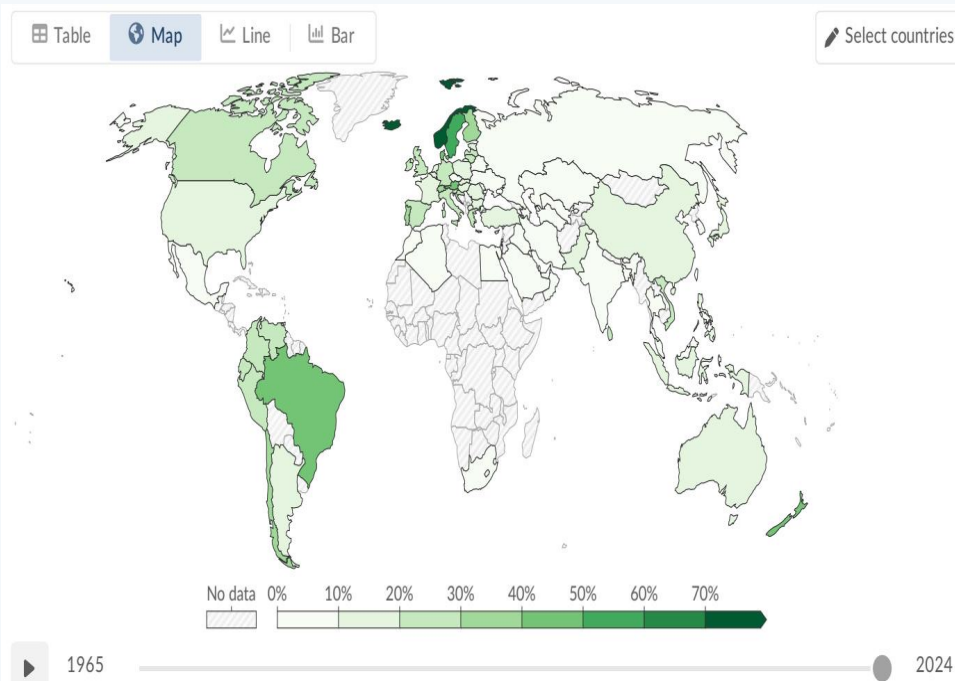
of renewable
electricity

Hydropower is mature and dispatchable, but expansion can be constrained by geography, drought risk, and environmental trade-offs.

Strategic role: stabilize high wind/solar grids with reservoir flexibility where available.

Wind and Solar Are the Fastest-Growing Renewable Generators

Combined generation now approaches hydropower levels

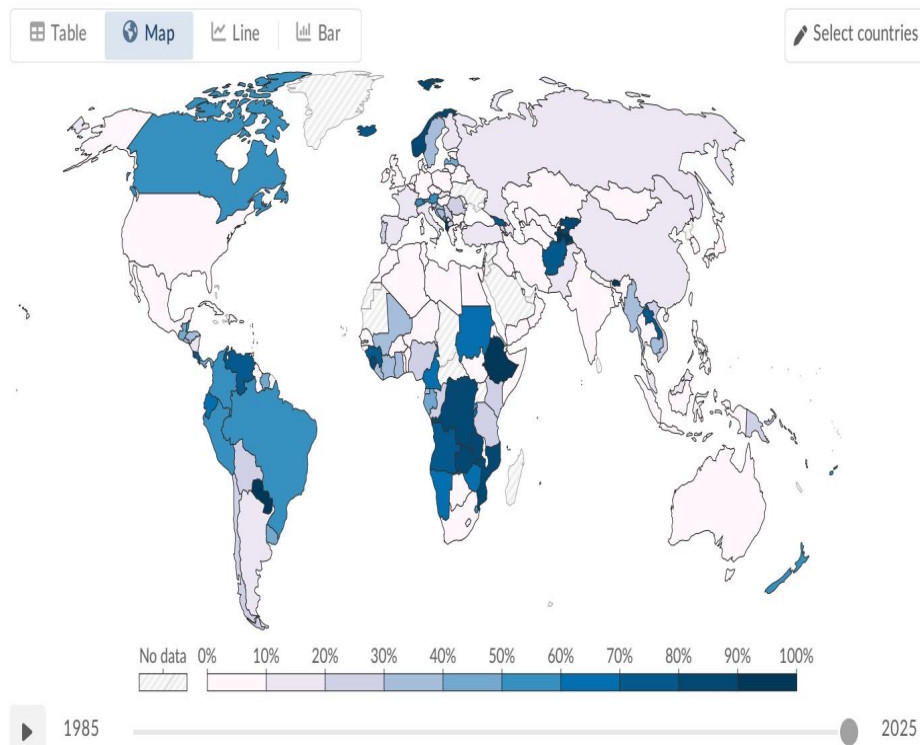


Wind: ~2,400 TWh (~25%) | Solar: ~2,000 TWh (~21%)

Solar is the faster riser from a lower base; wind remains larger in absolute output for now.

Renewable Transition Is Highly Uneven: Leaders Above 60%, Laggards Below 10%

Resource endowment and policy ambition shape outcomes



Scandinavia & Brazil

65–98% renewable electricity, anchored by hydropower.

Canada

~65% renewable electricity with strong hydro base.

China & India

Rapid wind/solar expansion, but lower shares (~35% and ~25%).

Middle East & parts of Africa

Many systems remain <10% due to fossil lock-in and underinvestment.

Key Takeaways: Progress Is Real, Pace Is Still Insufficient

Global Renewable Energy in 2025

- ~14% of global primary energy and ~30% of electricity now come from renewables.
- Hydropower remains dominant (~47% of renewable electricity), but wind and solar are rapidly converging.
- Wind (~2,400 TWh) and solar (~2,000 TWh) are the fastest-growing renewable generators.
- Regional disparities remain stark: leaders exceed 60–90% while many countries stay below 10%.
- Next phase requires accelerating grids, storage, finance, and transport/heating electrification.

Call to action: shift from incremental deployment to system-wide acceleration this decade.